# BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

RESEARCH / PROTOTYPE SYSTEM DISCLAIMER: This system is a research/prototype fingerprint-based blood-group classification system. It is not clinically validated and must not be used as a substitute for standard serological blood-group testing or for medical/transfusion decisions. All predictions and advisory indicators are provided solely for evaluation purposes.

## 1. EXAMINATION & SAMPLE AUDIT DETAILS

| | | | |
|---|---|---|---|
| **Report ID:** | BG-000596 | **Timestamp:** | 2026-09-08 21:41:17 |
| **Input Source:** | Mantra MFS100 Scanner \| Standard Finger: LEFT THUMB | **Evaluation Status:** | **NORMAL** |
| **Model Name:** | Champion Ensemble V2 (ResNet50 58% + VGG16 38% + EffNet 4%) | **Confidence:** | **94.93%** |
| **Standard Acquisition Finger:** | **LEFT THUMB** (Convention-based protocol — not automatically verified by scanner) | | |
| **Advisory Status:** | **CONFIDENCE OK** (τ=70%) | **Advisory Reason:** | **NONE** |

## 2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP



# Predicted Blood Group:

# O+

Confidence Score: **94.93%**
Advisory Threshold: **70%**
Advisory Screening: **CONFIDENCE OK**
Derived ABO Group: **O** (95.3% prob) | Rh Factor: **+**

**RESEARCH ADVISORY:** Advisory threshold passed (≥ 70.0%). Note: Confidence OK does not constitute medical verification. Standard serological testing required for medical/transfusion use.

## 3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

| A+ | A- | AB+ | AB- | B+ | B- | O+ | O- |
|---|---|---|---|---|---|---|---|
| 2.0% | 1.8% | 0.0% | 0.9% | 0.0% | 0.0% | 94.9% | 0.4% |